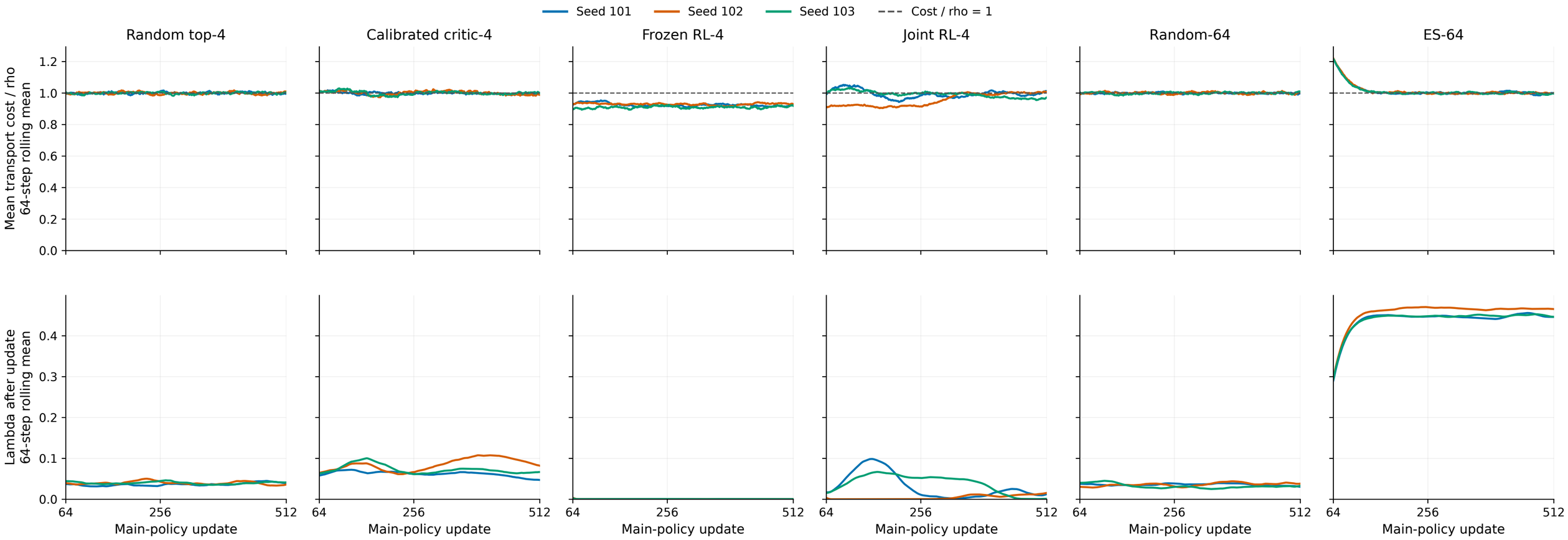

B training budget and dual-price trajectories (degree 30; 512 updates)
Seed 101
Seed 102
Seed 103
Cost / rho = 1
Random top-4
Calibrated critic-4
Frozen RL-4
Joint RL-4
Random-64
ES-64
Mean transport cost / rho
64-step rolling mean
Lambda after update
64-step rolling mean
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.1
0.3
64
256
512
Main-policy update
512-update histories; full 64-update windows plotted from update 64. Final evaluation uses the validation-selected best checkpoint.
The dashed line is an average transport-budget reference, not a per-step feasibility guarantee.